Chaeyoon Shin

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Education

Sungkyunkwan University (SKKU) — Seoul, Korea
M.S. in Digital Health, Samsung Advanced Institute for Health Sciences & Technology (SAIHST) — *Mar 2026 – Present*
- Advisor: Dr. Yoonjung Yoonie Joo

Sungkyunkwan University (SKKU) — Suwon, Korea
B.E. in Integrative Biotechnology (Magna Cum Laude) — *Mar 2022 – Feb 2026*

The University of Texas at Dallas — Richardson, TX
Fall Exchange Student Program — *Aug 2024 – Dec 2024*

Publications

Preprints

[1] **Chaeyoon Shin**, Junghoon Park, et al. Psychological resources confer greatest protection against late-life depression in genetically vulnerable adults. *Preprint*, Jun 2026. doi:10.21203/rs.3.rs-9990161/v1

[2] **Chaeyoon Shin**, Jinyun Kim, Se In Sung, Yoonjung Yoonie Joo. Heterogeneous associations of latency duration with neonatal outcomes after preterm prelabor rupture of membranes: a nationwide cohort study. *Preprint*, Sep 2026. doi:10.64898/2026.09.07.26361886

Projects

A Study on Establishing the Foundation for Linking and Collecting Patient-Generated Health Data (PGHD) in Republic of Korea National Bio Bigdata
Korea Health Information Service (KHIS) — *Nov 2025 – Oct 2026*
- Conducted an 11-month pilot study involving 187 participants recruited through Samsung Medical Center, collecting consumer wearable device data (Samsung Galaxy Watch) and continuous glucose monitoring (CGM) data to evaluate the feasibility of integration into BioBigData.Korea (BIKO)
- Established and applied quality-control criteria for wearable and CGM data based on the 4C1V framework (conformance, completeness, consistency, correlation, validity)
- Assessed participant adherence and data completeness, and proposed key wearable variables along with the minimum collection period and sample size for future large-scale collection
- Conducted an extensive literature review of PGHD collection in biobanks and population cohorts

Research Experience

Undergraduate Research Intern — Jan 2025 – Feb 2026
Genomics & Neural Network Intelligence for Healthcare (GENNIE) Lab, SKKU

Undergraduate Research Program (URP) — Dec 2023 – Aug 2024
Computational Cognitive Affective Neuroscience (COCOAN) Lab, SKKU
- Conducted fMRI data analysis and derived activation maps using MATLAB
- Participated in pilot studies and assisted fMRI experiments

Industry–Academia Collaboration Project — Jan 2024 – Feb 2024
Lab of Molecular Development and Cell Signaling, SKKU
- Trained in cell culture and experimental techniques in ovarian cancer research
- Performed JC-1 staining, hanging drop assays, and tissue preparation protocols

Conference Presentations

Poster Presentation, American Society of Human Genetics (ASHG) — Oct 2026
"Neuroticism polygenic score marks stronger protective associations between psychological resources and late-life depressive symptoms: evidence from a 35-year sociogenomic cohort"

Poster Presentation, Korean Society of Medical Informatics (KOSMI) — Jun 2026
"Interplay of Psychological and Genetic Factors in Shaping Late-Life Depression"
- Co-authored poster awarded **Best Poster**

Teaching & Mentoring

Teaching Assistant, *Introduction to Multiomics Analysis* — Fall 2026

Teaching Assistant, *Introduction to Healthcare Data Analytics* — Spring 2026

Awards & Honors

Scholarships

SKKU Graduate Scholarship (100% tuition support for Master's Program) — 2026

KorUS Educational Exchange Initiative for Youth in STEM ($9,000/semester), issued by KIAT — 2024

SKKU Next Generation Scholars Program (100% tuition support for 3rd–4th undergraduate years) — 2024

SKKU Academic Excellence Scholarship (top 10%) — 2023

Extracurriculars

Member, SKKAI (Sungkyunkwan AI Association), Medical AI Team — Sep 2025 – Jan 2026

- Explainable AI (XAI) research (sparse autoencoders) on pathology foundation models

Member, National Undergraduate Symposium on Biology — Aug 2023 – Sep 2023

- Delivered a group presentation on cellular senescence and immunometabolism

Skills

Languages: English (TOEFL 107/120, TOEIC 960/990), Korean (Native)
Programming: R, Python, Linux/Bash